

Spanning a range of voltages and temperatures

With power-conversion components present in all kinds of electronic devices/gadgets, it is imperative for these to support a range of temperature and voltage operations. Components are rated according to grids. Commercial applications are the ones working at least power and temperature ranges, but have to take into account power fluctuations. Ones used under the hood of automobiles and light emitting diodes (LEDs) need to support about 150°C temperature. Reliability and high tolerance are prime deciders for military-grade components, while those that go into space electronics need to withstand many kinds of radiations.

Products that work on AC mains like LED bulbs and inverters take in 220V AC input and convert it to high-range voltages. High voltages demand heavy device capability, and to support these, there are MOSFETs, bipolar transistors and insulated-gate bipolar transistors (IGBTs), competing in the same space.

Traditionally, voltage handling was highest for bipolar technology. With IGBT disrupting this technology, it is now common to find IGBT modules in high-voltage equipment like the uninterrupted power supply (UPS).

Batteries to the rescue. Battery-

Harvest energy and use it efficiently

Take the case of a sensor that is fitted into a wall. It is left untouched, but expected to do its job sincerely, day in and day out. Would someone have to monitor its battery levels regularly? Instead, put a small board next to it and fit in a solar panel, not unlike modern electronics calculators. The sun would then take care of keeping the battery charged.

Using brushless DC motors in appliances like fans, variable speed drives for controlling motors in air-conditioners and refrigerators, and employing smartmeters in smartgrids helps reduce energy loss and conserve energy.

Further, renewable energy power supplies such as solar/electric vehicles and telecom/server power supplies demand high efficiencies of nearly 99 per cent. Energy harvesting also comes into utmost importance when dealing with the Internet of Things applications.

Semitrex Technologies, a Californian start-up, claims to have come up with one-size-fits-all power supply on a chip. Tronium, which can provide an output from 1.8V to 48V. The design of the chip is based on a capacitive-voltage-reduction technology termed muxcapacitor, and it completely does away with magnetic and inductive components.



Tronium power supply SoC (Image courtesy: www.semitrex.com)

powered devices like mobile phones and wearables perform power conversion at direct current (DC) range, and at much lower levels. Battery is expected to charge fast, be stable and reliable. A power supply also needs to tolerate high voltages and not break down in such scenarios. To add to the burden, there is the case of wireless charging.

Many battery-operated systems use two or more supercapacitors in series. In an earlier interview,

Robert L. Chao, founder, Advanced Linear Devices Inc., explained how MOSFET based current balancing is enabling reduced power use in superconductor stacks. He says, "A MOSFET array balances leakage currents in each cell by exponentially varying currents relative to selected operating voltages. This method compensates for leakage-current imbalances with small voltage imbalances so that maximum rated voltage limits are not exceeded."

Varying models for different end applications

While designing power-conversion components, lifestyle of consumers and application of the system also define scalability of the design. Current-handling capability and failure protection of the designed circuit reflects its complexity, cost and size.

Consider the case of a metering device. Presently, a majority of digital residential meters are non-communicative, with no electronic controlling capabilities. Power requirement for such systems will hardly be 0.5W to 2W. However, futuristic smartmeters look to integrate communication capabilities and remote control facilities within, and this would demand a difference in the design. Thus, system developers design variant models of power-sourcing circuits.

Conquering heat

Small sizes and higher packing densities automatically come with

Is AC losing out to DC?

Most appliances used in daily life have a rectifier circuit that converts supplied AC current into direct current (DC), and store energy or use it to power the system. With solar and battery-powered technologies gaining importance, DC-to-DC conversion is becoming the norm. Output of a solar photovoltaic system is low-voltage DC that can be boosted to suit requirements. These are highly efficient and preferred. It is in the case of home lighting systems that AC is still preferred, as AC loads with high power rating are easily available.

High-voltage DC transmission (HVDC) is also a growing trend, showing promise in improving overall system efficiency and renewable energy opportunities. In a white paper, Stephen Oliver, vice president, VI Chip Product Line, Vicor Corp., has said that, innovative conversion, control and distribution approaches in power-conversion technology, enabled by advanced semiconductors and conversion topologies are making HVDC practical for distribution.

HVDC offers tangible and significant benefits for both sourcing options and system end-to-end performance. A research by France Telecom and China Mobile estimates that DC distribution can save eight per cent to ten per cent of power across the board.